STRUCTURAL SENSITIVITY IN A SPIKING NEURAL NETWORK

An independent reader’s guide to the molecular graph-intervention research, accompanying the full scientific article and verifiable computational records.

Evidence snapshot: 23 October 2026 | Computational study, not a clinical investigation

What does the research ask?

Can the activity of a deterministic spiking neural network distinguish selected changes in the connections of computational molecular graphs? Do learning and recurrent feedback change that sensitivity? The study tests these questions by intervening on graph connections and measuring the resulting native network activity.

What is the system?

The underlying compiled neural scaffold contains 118 neurons and 324 directed synapses. An independently reset network runs 48 deterministic ticks per simulation. Molecular representations provide declared computational connections to this system; altering those connections creates matched comparison pairs.

What was actually measured?

Graph-label classification — A frozen classifier tries to distinguish intact versus rewired computational graph labels using held-out native spike responses.

Neural firing-mask disagreement — The proportion of 48 × 22 recorded target-neuron positions that change between a matched pair of native executions.

Algorithmic molecular representations — RNA nested-pair predictions, limited energy-model calculations, RDKit 3D conformers and MMFF94 short-fragment minimization supply later inputs; they are not biological outcome measurements.

Main controlled graph result

Intervention	Correct labels	Computed accuracy
Cycle-breaking graph	196/256	76.56%
Separate cycle-preserving control	116/256	45.31%

The two values refer to different interventions on a constructed graph associated with PDB identifier 1KF1. They do not measure the accuracy of DNA folding, nor do they prove general graph equivalence recognition.

HOW THE EVIDENCE IS ORGANIZED

The chain from method to claim

A researcher should be able to follow each reported value from computational input and intervention, through archived native masks, to a source-derived table or figure. The publication includes 13,312 audited paired records, 208 condition-level summary rows, 100 historical classifier rows and 14 source-derived figures.

Why negative controls matter

The additional RNA-labelled classification cases did not pass the originally recorded multiple-test correction. The cycle-preserving structural contrast was not reliably discriminated. Under one computational spatial-contact definition, ethanol had no qualifying nonbonded contacts and correspondingly no contact-ablation response. In some cases frozen learning produced larger neural differences than learning-enabled settings. These results help prevent an overstated interpretation.

Chemical and RNA computations

The chemical experiments process real SMILES-encoded molecular graphs. Subsequent methods derive RNA base-pair contacts and three-dimensional chemical contacts from computational algorithms. The atomic-force study uses MMFF94 on eight small molecules and five four-nucleotide RNA fragments; it does not optimize complete RNA folds. Algorithmic model energies are not measured experimental energies.

Verification and reproducibility

The latest release retains original data, named tests, logs, publication figures, source ZIPs, a reader-readable data dictionary, and a mapping that verifies every original file’s SHA-256 after directory reorganization. The 81 named final tests are preserved historical results, not reruns performed to create this guide.

How this connects to scientific publication

A research article presents the question, hypotheses, methods, results and interpretation. A Data Descriptor would instead focus on dataset generation and technical validation. Editorial preparation additionally requires real authorship, contribution, funding, competing-interest and AI-assistance disclosures, plus any venue-specific deposit requirements. None of these are fabricated in the companion package.

Start reading

Begin with 01_Research_Article/Research_Article.pdf. Examine 02_Publication_Figures/ for the 14 source-derived figures, 04_Data_And_Results/ for numerical tables, and 07_Reproduce_And_Verify/ for verification instructions. The preserved original source release is under 08_Executable_Source_Archive/.

This guide explains the current study; it does not substitute for the source, experimental methods, or independent biological validation.